Everything You Wanted to Know About Quality

Plain-language definitions of the specifications, quality terms and inspection language you'll see on drawings and purchase orders for precision laser welding. Use this as a reference; your drawing, purchase order and the governing specification always take precedence.

Specifications and programs

A summary and key points for each specification and program you're likely to see on a laser welding job. Where a spec is free, we link the full document. Copyrighted standards are sold only by their publishers, so we link to the official source.

WS-LWS-001, Weld Solutions Laser Welding Specification

Our internal specification for laser welding. It defines how we develop, document, qualify and inspect every laser weld we make. Current revision: Rev B.

- Covers weld development, weld schedules, procedure qualification and operator qualification
- Rev B aligns procedure and operator qualification with AWS D17.1
- Visual weld inspection is performed at 10X magnification
- Applies to all Weld Solutions laser welding unless your drawing or PO calls out a different spec

Full spec free on request: quality@weldsolutions.com

AWS D17.1/D17.1M, Fusion Welding for Aerospace Applications

The American Welding Society specification for welding aerospace hardware, and the spec most often called out for laser welding in aerospace and defense. Current edition: 4th edition, 2024.

- Applies to fusion processes including laser beam and electron beam welding
- Requires a qualified welding procedure and qualified welders or operators
- Welds are assigned Class A, B or C. Class A is the most critical and has the tightest acceptance limits
- Defines inspection methods and acceptance criteria for each class
- Replaced the older MIL-STD-2219 and is widely used in place of MIL-STD-1595 for qualification

Purchase from AWS: https://pubs.aws.org/

MIL-STD-1595 / AMS-STD-1595, Qualification of Aircraft, Missile and Aerospace Fusion Welders

The long-standing military standard for qualifying aerospace welders and welding operators.

- A qualification standard for people, not a weld procedure or acceptance spec
- Transferred to SAE and continues as AMS-STD-1595
- Many aerospace customers now qualify welders and operators to AWS D17.1 instead
- Still seen on older drawings and legacy programs

Purchase from SAE: https://www.sae.org/standards/

MIL-STD-2219, Fusion Welding for Aerospace Applications

The former military standard for fusion welding of aerospace hardware.

- AWS D17.1 was developed to replace it
- Older drawings that call it out are commonly worked to D17.1 with customer approval

Free historical copy from DLA ASSIST: https://quicksearch.dla.mil/

AWS C7.4/C7.4M, Process Specification and Operator Qualification for Laser Beam Welding

An AWS specification written specifically for laser beam welding. Current edition: 2017.

- Defines what a laser welding process specification must contain
- Defines how laser welding operators are qualified
- Useful for laser work outside aerospace, where D17.1 may not apply

Purchase from AWS: https://pubs.aws.org/

SAE AMS 2680, Electron-Beam Welding for Fatigue Critical Applications

An SAE Aerospace Material Specification for electron beam welding where fatigue life is critical, such as rotating engine hardware.

- Tighter controls than general welding specifications on procedure, inspection and acceptance
- Written for electron beam welding; some customers apply it to laser welding by agreement
- Companion to AMS 2681

Purchase from SAE: https://saemobilus.sae.org/content/AMS2680C

SAE AMS 2681, Electron-Beam Welding

The general SAE specification for electron beam welding of aerospace parts that are not fatigue critical.

- Companion to AMS 2680 for non-fatigue-critical applications
- Covers procedure, operator qualification, inspection and acceptance

Purchase from SAE: https://saemobilus.sae.org/content/AMS2681A

ISO 9001, Quality Management Systems

The international standard for a quality management system in any industry. Current edition: ISO 9001:2015, with a 2024 amendment.

- Documented processes, records and management review
- Customer focus and risk-based thinking
- Control of nonconforming product and corrective action
- Calibration, training and supplier control
- Continual improvement
- The foundation that ISO 13485 and AS9100 build on

Purchase from ISO: https://www.iso.org/standard/62085.html

ISO 13485, Medical Devices Quality Management Systems

The quality management standard for medical device manufacturers and their suppliers. Current edition: ISO 13485:2016.

- Built on ISO 9001, with more emphasis on documentation and regulatory compliance
- Requires validation of processes that can't be fully verified by inspection, which includes welding
- Risk management throughout the product life cycle
- Strong traceability and record retention requirements
- Now incorporated into U.S. FDA regulation through the QMSR

Purchase from ISO: https://www.iso.org/standard/59752.html

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21 CFR Part 820, Quality Management System Regulation (QMSR)

The U.S. FDA regulation governing quality systems for medical device manufacturers.

- Effective February 2, 2026, it incorporates ISO 13485:2016 by reference
- Replaced the former Quality System Regulation (QSR)
- Applies directly to device manufacturers; suppliers see it through purchase order flowdowns
- Free to read in full from the U.S. government

Download full regulation free (eCFR): https://www.ecfr.gov/current/title-21/chapter-I/subchapter-H/part-820

AS9100, Aerospace Quality Management Systems

The quality management standard for aviation, space and defense. AS9100D is the current revision; the IAQG is replacing it with the IA9100 series.

- Built on ISO 9001, with added aerospace requirements
- Configuration management and first article inspection
- Control of special processes such as welding
- Counterfeit part prevention and product safety
- On-time delivery and supplier performance monitoring
- Certification is granted by accredited third-party registrars

Purchase from SAE: https://www.sae.org/standards/content/as9100d/

AS9102, Aerospace First Article Inspection

The aerospace standard that defines how a first article inspection is performed and documented.

- Form 1: part number accountability
- Form 2: product accountability, covering materials and special processes
- Form 3: characteristic accountability, verifying every dimension and note
- Required on new parts and after changes to design, process or location

Purchase from SAE: https://www.sae.org/standards/

Nadcap

An industry-managed accreditation program for special processes in aerospace, administered by the Performance Review Institute.

- Covers processes such as welding, heat treating, chemical processing and NDT
- Many aerospace primes require it for suppliers welding their hardware
- Welding audits follow the AC7110 audit criteria
- Audits are performed by industry auditors and reviewed by a task group of primes and suppliers
- Separate from AS9100: AS9100 covers the quality system, Nadcap covers the special process itself

Program information: https://p-r-i.org/nadcap/

IQ, OQ, PQ, Process Validation

The three stages used to prove equipment and a process are capable, done in order. Required by most medical device customers and increasingly by aerospace.

- IQ, Installation Qualification: equipment is installed to specification, with utilities, calibration and documentation in place
- OQ, Operational Qualification: the process is challenged at the upper and lower limits of its parameter window and still makes good welds
- PQ, Performance Qualification: the process makes good parts consistently under normal production conditions, over multiple runs
- Each stage has a written protocol with acceptance criteria and a signed report
- Changes to the equipment or process may require revalidation

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ISO 13919-1, Quality Levels for Electron and Laser Beam Weld Imperfections

Defines acceptance levels for imperfections in electron and laser beam welds in steel.

- Quality levels B, C and D, with B the most stringent
- Common on European and international drawings

Purchase from ISO: https://www.iso.org/

ISO 15614-11, Welding Procedure Test for Electron and Laser Beam Welding

The ISO method for qualifying an electron or laser beam welding procedure by testing welded samples.

- The international counterpart to a PQR under AWS specifications

Purchase from ISO: https://www.iso.org/

Glossary: qualification and validation

How a weld process is proven before production parts are made.

Process validation

Documented evidence that a process will consistently produce a result meeting its requirements. Required whenever the result cannot be fully verified by later inspection, which is the case for welding.

Special process

A process whose quality cannot be fully confirmed by inspecting the finished part, so the process itself must be qualified and controlled. Welding, heat treating, plating and NDT are typical special processes.

WPS, Welding Procedure Specification

The written instructions for making a specific weld: material, joint design, equipment, laser parameters, shielding gas, fixturing and inspection. The operator works to the WPS.

PQR, Procedure Qualification Record

The record of the test welds and test results that prove a WPS produces acceptable welds. A WPS is qualified by its supporting PQR.

Operator qualification

Documented proof that a specific operator can produce acceptable welds with a specific process, usually by welding and testing qualification samples. Laser and electron beam welders are typically qualified as operators of mechanized or automatic equipment. Qualifications have defined limits and must be maintained.

Essential variable

A parameter that, if changed beyond the limits in the governing specification, requires the procedure to be requalified. Examples include material type, joint design, laser type and significant changes in energy input.

Weld schedule

The set of machine parameters used for a particular part and joint, such as pulse energy, pulse width, frequency, spot size, travel speed and gas flow.

Weld development

The trials used to establish a weld schedule for a new part, normally confirmed with cross sections before production.

FAI, First Article Inspection

A complete inspection of the first part, or first lot, from a new or changed process to confirm that every drawing requirement is met before production continues.

Golden sample

An approved reference part, kept as the visual and dimensional standard that production parts are compared against.

Glossary: quality system terms

The paperwork and controls behind the parts.

QMS, Quality Management System

The documented set of policies, procedures and records a company uses to control quality. ISO 9001, ISO 13485 and AS9100 each define requirements for a QMS.

Flowdown

Requirements passed from a customer's contract or purchase order to the supplier, and from the supplier to its own subcontractors.

CoC, Certificate of Conformance

A signed statement from the supplier that the parts or services furnished meet the purchase order and drawing requirements.

Material certification, MTR

The mill test report or certificate of analysis that documents the chemistry and properties of a material lot, tied to a heat or lot number.

Traceability

The ability to trace a part back through its records: material lot, operator, equipment, weld schedule, date and inspection results.

Lot

A quantity of parts processed together under the same conditions and tracked as a group.

Traveler or router

The document that travels with a job through each step, recording who did what, when, and with which equipment and parameters.

Calibration

Comparing a measuring or process instrument against a traceable standard and adjusting or recording its accuracy at defined intervals.

Nonconformance and NCR

Any product or process that does not meet a requirement. The Nonconformance Report (NCR) documents it, isolates the affected parts and records the disposition.

MRB, Material Review Board

The group, often including the customer, that decides the disposition of nonconforming product: rework, repair, use as is, return to vendor or scrap.

Deviation and waiver

Written customer approval to depart from a requirement. A deviation is normally approved before the work is done; a waiver accepts product after the fact.

CAPA, Corrective and Preventive Action

The formal process for finding the root cause of a problem, fixing it, preventing recurrence and verifying the fix worked.

Root cause analysis

Methods such as 5 Whys and fishbone (Ishikawa) diagrams used to find the underlying cause of a problem rather than its symptom.

FMEA and PFMEA

Failure Mode and Effects Analysis. A structured review of what could go wrong, how severe it would be, how likely it is, and how it would be detected. A PFMEA applies this to a manufacturing process.

Control plan

A document listing the process controls, inspections, frequencies and reaction plans used to keep a process producing good parts.

SPC, Statistical Process Control

Using control charts to monitor a process so that drift is caught before bad parts are made.

Cpk

A process capability index showing how well a process fits within its tolerance and how well it is centered. A Cpk of 1.33 or higher is a common minimum.

MSA and Gage R&R

Measurement Systems Analysis. A Gage Repeatability and Reproducibility study shows how much of the observed variation comes from the measuring method and the people using it.

DHR, DMR, DHF

Medical device records. The Device History Record (DHR) documents how each lot was built. The Device Master Record (DMR) is the complete recipe for the device. The Design History File (DHF) documents the design. Under the QMSR the Medical Device File replaces the DMR terminology.

Objective evidence

Records that show a requirement was met: inspection data, signed travelers, certifications, photos and cross sections.

Glossary: inspection and testing

How welds are checked.

Visual inspection

Examining the weld surface for cracks, porosity, undercut, spatter, discoloration and size. At Weld Solutions, visual weld inspection is performed at 10X magnification.

Destructive and nondestructive testing

Destructive testing cuts, pulls or bends a sample to evaluate the weld and destroys it. Nondestructive testing (NDT) evaluates a weld without damaging the part.

Metallographic cross section (metlab)

Cutting through a weld, mounting, polishing and etching it, then examining it under magnification to measure penetration, weld width and internal defects.

Macro and micro examination

Examination of an etched cross section at low magnification (macro) for weld shape and penetration, or high magnification (micro) for grain structure and fine defects.

Penetration depth

How far the fused metal extends into the joint, measured on a cross section. A key acceptance criterion for most laser welds.

Effective weld depth and throat

The portion of the weld that actually carries load, used in strength and acceptance calculations.

Radiographic testing (RT)
X-ray inspection used to find internal porosity, cracks and lack of fusion.

Penetrant testing (PT)
A dye or fluorescent penetrant inspection that reveals surface-breaking cracks.

Leak testing and hermeticity
Testing a sealed assembly to confirm it does not leak. Helium mass spectrometer leak testing is the common method for hermetic laser-sealed packages, with results reported as a leak rate.

Witness sample
A test piece welded with production parts, using the same setup, then sectioned or tested to verify the process without destroying a production part.

Pull test and shear test
Mechanical tests that load a weld to failure to confirm strength.

Glossary: weld discontinuities

What inspectors look for. A discontinuity becomes a defect only when it exceeds the acceptance criteria of the governing specification.

Porosity
Gas pockets trapped in the weld metal. Can be surface or internal.

Crack
A fracture in the weld or heat affected zone. Typically rejectable in any size under aerospace and medical specifications.

Lack of fusion
Weld metal that did not fuse to the base metal or to previous weld metal.

Incomplete penetration
Weld that did not reach the required depth in the joint.

Undercut
A groove melted into the base metal along the edge of the weld and left unfilled.

Underfill and concavity
A weld face that sits below the surface of the adjoining base metal.

Burn-through
Excess energy that melts completely through the material, leaving a hole.

Spatter
Droplets of metal expelled during welding that land on the part.

Crater
A depression at the end of a weld bead. Controlled in laser welding with power ramp down.

Discoloration and heat tint
Oxide color on and near the weld caused by inadequate shielding. Some specifications limit the allowable colors.

Mismatch
Misalignment between the two parts at the joint before welding.

Glossary: laser welding terms

The process language behind the parameters.

Conduction mode

Laser welding where heat flows into the part by conduction, giving a shallow, wide, smooth weld. Used for thin parts and cosmetic seams.

Keyhole mode

Laser welding at high power density that vaporizes a narrow channel in the metal, giving deep, narrow penetration.

Pulsed laser welding

The laser fires in discrete pulses. Each pulse makes a spot weld; overlapping spots make a seam. Nd:YAG lasers are typically pulsed.

Continuous wave (CW)

The laser emits continuously while the beam or part moves, producing a continuous seam. Fiber lasers are commonly run CW or modulated.

Pulse energy

The energy of a single pulse, in joules.

Peak power

The power during the pulse, in watts or kilowatts. Pulse energy divided by pulse width.

Pulse width

The duration of each pulse, in milliseconds.

Frequency

The number of pulses per second, in hertz.

Average power

Pulse energy multiplied by frequency, in watts.

Spot size

The diameter of the focused beam on the part.

Overlap

The percentage each pulsed spot overlaps the previous one. Higher overlap gives a smoother, more continuous seam. Hermetic seams typically need high overlap.

Wobble welding

Oscillating the beam in a small pattern as it travels, to widen the weld, bridge gaps and stabilize the weld pool.

Shielding gas

Inert gas, usually argon, that protects the molten weld from oxidation.

HAZ, Heat affected zone

Base metal next to the weld whose properties are changed by heat but which did not melt. Laser welding produces a very small HAZ.

Fiber laser and Nd:YAG laser

Two common industrial laser sources. Both operate near 1,064 nm wavelength; fiber lasers offer high efficiency and beam quality, while lamp-pumped Nd:YAG lasers are widely used for pulsed precision and jewelry welding.

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